

A DOA Estimation Method for Distributed Nested Arrays Based on Non-Circular Signals

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Abstract—To address the requirements of high accuracy and unambiguous estimation in distributed array direction-of-arrival (DOA) estimation, a three-level ambiguity resolution DOA estimation method based on non-circular signals for distributed nested arrays is proposed. First, the conjugate symmetry property of non-circular signals is exploited to construct an extended covariance matrix, which is vectorized and rearranged to form five virtual subarrays located at 0 , $\pm B$, and $\pm 2B$. Subsequently, a weighted eigensubspace algorithm is employed to perform spectral estimation for each virtual subarray sequentially, implementing a coarse-to-fine three-level ambiguity resolution strategy. The non-circular phase consistency is innovatively introduced as an auxiliary criterion to enhance the robustness of ambiguity resolution. Simulation results demonstrate that the proposed method can effectively exploit long baselines to improve estimation accuracy while ensuring unambiguous estimation.

Index Terms—DOA estimation, distributed nested array, non-circular signal, ambiguity resolution

I. INTRODUCTION

Distributed arrays overcome aperture limitations of conventional sparse arrays by spatially deploying multiple subarrays, achieving equivalent baseline lengths far exceeding those of single arrays and enabling superior angular measurement accuracy [1]. Nevertheless, when the inter-subarray baseline exceeds half the signal wavelength, periodic grating lobes emerge in the array pattern, inducing DOA estimation ambiguity [2]. This grating lobe ambiguity constitutes a critical bottleneck impeding practical deployment.

To address this challenge, multi-baseline progressive ambiguity resolution methods [3] exploit spatial spectral characteristics of varying baseline lengths through coarse-to-fine search strategies. Joint DOA estimation approaches based on rotational invariance [4] leverage inter-subarray phase differences for unambiguous estimation. Additionally, sparse reconstruction methods [5] reformulate ambiguity resolution as sparse optimization problems. However, these methods predominantly target circular signals and fail to exploit the sum-difference co-array properties of non-circular signals.

Non-circular signals, characterized by non-zero conjugate correlation, include typical modulations such as BPSK, ASK, and AM [6]. Unlike circular signals, both the covariance and conjugate covariance matrices of non-circular signals contain

effective DOA information. Reference [7] demonstrates that for non-circular signals, a sum co-array can be constructed alongside the difference co-array, forming a sum-difference co-array that significantly increases virtual elements. References [8], [9] validate the superiority of this structure in coprime and nested arrays. Recent studies have explored optimized sparse nested arrays for non-circular signals with enhanced DOFs [10], as well as off-grid sparse Bayesian learning approaches [11].

Integrating non-circular signal properties with distributed nested arrays simultaneously captures the DOF advantages of sum-difference co-arrays and the high-accuracy benefits of long baselines. The extended covariance matrix enables formation of multiple virtual subarray groups with different baseline lengths, providing a natural hierarchical structure for multi-level ambiguity resolution. However, research on DOA estimation and ambiguity resolution for distributed nested arrays exploiting non-circular signal sum-difference co-arrays remains limited.

This paper proposes a three-level ambiguity resolution DOA estimation method for distributed nested arrays based on non-circular signals. The main contributions include: (1) establishing a non-circular signal extension model that constructs five virtual subarrays at positions 0 , $\pm B$, and $\pm 2B$ through covariance matrix vectorization; (2) designing a three-level ambiguity resolution strategy that achieves progressive refinement from coarse to fine estimation; (3) introducing non-circular phase consistency as an auxiliary criterion to eliminate false peaks at grating lobe positions.

II. SIGNAL MODEL

Consider a distributed nested array structure consisting of two identical nested subarrays, with a baseline length of B between the phase centers of the subarrays (where $B \gg \lambda/2$, and λ is the signal wavelength). For convenience, B is assumed to be an integer multiple of $\lambda/2$. Each nested subarray is composed of two cascaded uniform linear arrays (ULAs): the first ULA contains N_1 elements with inter-element spacing $d_1 = \lambda/2$; the second ULA contains N_2 elements with inter-element spacing $d_2 = (N_1 + 1)d_1$. For convenience of analysis, let $N_1 = N_2 = N$, then the total number of elements in a

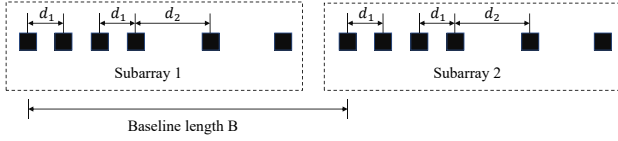


Fig. 1. Array configuration of distributed nested array.

single nested subarray is $M = 2N$, and the total number of elements in the distributed array is $2M = 4N$.

Assume P far-field narrowband non-circular signals impinge on the array from directions $\theta_1, \theta_2, \dots, \theta_P$. The distinctive feature of non-circular signals is that their conjugate correlation matrix is non-zero, which can be characterized by introducing the non-circular phases $\phi_1, \phi_2, \dots, \phi_P$. Specifically, non-circular signals can be modeled as $s_p(t) = e^{j\phi_p} \cdot r_p(t)$ for $p = 1, 2, \dots, P$, where $r_p(t)$ is a real-valued random process, and $\phi_p \in [0, 2\pi)$ is the non-circular phase of the p -th source. This model encompasses typical non-circular modulated signals such as BPSK and ASK.

At time instant t , the received signals of the two nested subarrays are expressed as $\mathbf{x}_1(t) = \mathbf{A}_1 \Phi \mathbf{s}(t) + \mathbf{n}_1(t)$ and $\mathbf{x}_2(t) = \mathbf{A}_2 \Phi \mathbf{s}(t) + \mathbf{n}_2(t)$, where $\mathbf{s}(t) = [s_1(t), s_2(t), \dots, s_P(t)]^T$ is the source vector, $\Phi = \text{diag}(e^{j\phi_1}, e^{j\phi_2}, \dots, e^{j\phi_P})$ is the non-circular phase matrix and $\mathbf{n}_1(t)$ and $\mathbf{n}_2(t)$ are mutually independent additive white Gaussian noise vectors with variance σ_n^2 . The steering matrix of the first subarray is $\mathbf{A}_1 = [\mathbf{a}_1(\theta_1), \mathbf{a}_1(\theta_2), \dots, \mathbf{a}_1(\theta_P)]$, where $\mathbf{a}_1(\theta_p) = [e^{j\gamma_{1p}}, e^{j\gamma_{2p}}, \dots, e^{j\gamma_{Mp}}]^T$ with $\gamma_{mp} = 2\pi d_m \sin \theta_p / \lambda$ and d_m denoting the position of the m -th element. The steering matrix of the second subarray is $\mathbf{A}_2 = \mathbf{A}_1 \Psi$, where $\Psi = \text{diag}(e^{j\psi_1}, e^{j\psi_2}, \dots, e^{j\psi_P})$ is the diagonal phase matrix with $\psi_p = 2\pi B \sin \theta_p / \lambda$ reflecting the phase difference between the two subarrays.

Exploiting the properties of non-circular signals, the extended signal vector is constructed as

$$\mathbf{x}_{\text{ext}}(t) = \begin{bmatrix} \mathbf{x}_1(t) \\ \mathbf{x}_2(t) \\ \mathbf{x}_1^*(t) \\ \mathbf{x}_2^*(t) \end{bmatrix} = \begin{bmatrix} \mathbf{A}_1 \Phi \\ \mathbf{A}_2 \Phi \\ \mathbf{A}_1^* \Phi^* \\ \mathbf{A}_2^* \Phi^* \end{bmatrix} \mathbf{s}(t) + \begin{bmatrix} \mathbf{n}_1(t) \\ \mathbf{n}_2(t) \\ \mathbf{n}_1^*(t) \\ \mathbf{n}_2^*(t) \end{bmatrix} = \bar{\mathbf{A}} \mathbf{s}(t) + \mathbf{n}_{\text{ext}}(t) \quad (1)$$

where $(\cdot)^*$ denotes the complex conjugate operation, and $\bar{\mathbf{A}}$ is the $4M \times P$ extended steering matrix.

The extended covariance matrix is defined as

$$\mathbf{R}_{\text{ext}} = E[\mathbf{x}_{\text{ext}}(t) \mathbf{x}_{\text{ext}}^H(t)] = \bar{\mathbf{A}} \mathbf{R}_s \bar{\mathbf{A}}^H + \sigma_n^2 \mathbf{I}_{4M} \quad (2)$$

where $\mathbf{R}_s = E[\mathbf{s}(t) \mathbf{s}^H(t)] = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_P^2)$ is the source covariance matrix (assuming mutually independent sources), and $(\cdot)^H$ denotes the conjugate transpose. In practice, the covariance matrix is estimated from L snapshots:

$$\hat{\mathbf{R}}_{\text{ext}} = \frac{1}{L} \sum_{t=1}^L \mathbf{x}_{\text{ext}}(t) \mathbf{x}_{\text{ext}}^H(t) = \frac{1}{L} \mathbf{X}_{\text{ext}} \mathbf{X}_{\text{ext}}^H \quad (3)$$

Vectorizing the extended covariance matrix, we obtain

$$\mathbf{z} = \text{vec}(\hat{\mathbf{R}}_{\text{ext}}) = (\bar{\mathbf{A}}^* \odot \bar{\mathbf{A}}) \mathbf{p} + \sigma_n^2 \text{vec}(\mathbf{I}_{4M}) \quad (4)$$

where $\mathbf{p} = [\sigma_1^2, \sigma_2^2, \dots, \sigma_P^2]^T$ is the source power vector and $\odot$ denotes the column-wise multiplication operation. The elements of vector $\mathbf{z}$ correspond to the virtual array manifold, and the virtual element positions can be revealed by analyzing the structure of $\bar{\mathbf{A}}^* \odot \bar{\mathbf{A}}$.

Based on the non-circular signal extension model, the virtual element positions corresponding to the vectorized covariance data can be expressed as combinations of difference and sum sets of the original element positions. Let the element position sets of subarrays 1 and 2 be $\mathcal{S}_1 = \{s_{1,m}\}_{m=1}^M$ and $\mathcal{S}_2 = \{s_{1,m} + B/d_1\}_{m=1}^M$, respectively. The virtual element positions can be categorized into five types:

(1) Virtual subarray at position 0: The virtual element positions of this subarray are formed by the union of the difference co-array and sum co-array:

$$\mathcal{V}_0 = \{\pm(s_{1,i} - s_{1,j})\}_{i,j=1}^M \cup \{\pm(s_{1,i} + s_{1,j})\}_{i,j=1}^M \quad (5)$$

Extract the maximum contiguous uniform portion centered at the origin from $\mathcal{V}_0$, with N_0 elements, corresponding to virtual ULA positions $\{-(N_0 - 1)/2, \dots, -1, 0, 1, \dots, (N_0 - 1)/2\}d_1$. The signal model of this subarray can be expressed as

$$\mathbf{z}_0 = \mathbf{A}_0 \mathbf{p} + \sigma_n^2 \bar{\mathbf{e}}_0 \quad (6)$$

where $\mathbf{A}_0$ is the steering matrix of the virtual subarray, and $\bar{\mathbf{e}}_0$ is a vector with 1 at the center position and 0 elsewhere.

(2) Virtual subarray at position $+B$: The virtual element positions of this subarray are

$$\mathcal{V}_{+B} = \{(s_{1,i} - s_{1,j}) + B/d_1\}_{i,j=1}^M \cup \{(s_{1,i} + s_{1,j}) + B/d_1\}_{i,j=1}^M \quad (7)$$

Extracting the maximum contiguous uniform portion with N_{B+} elements, the signal model is

$$\mathbf{z}_{+B} = \mathbf{A}_{B+} \Psi \mathbf{p} \quad (8)$$

where $\Psi = \text{diag}(e^{j\psi_1}, e^{j\psi_2}, \dots, e^{j\psi_P})$, $\psi_p = 2\pi B \sin \theta_p / \lambda$.

(3) Virtual subarray at position $-B$: Similarly, the virtual element positions are

$$\mathcal{V}_{-B} = \{(s_{1,i} - s_{1,j}) - B/d_1\}_{i,j=1}^M \cup \{-(s_{1,i} + s_{1,j}) - B/d_1\}_{i,j=1}^M \quad (9)$$

With N_{B-} elements in the maximum contiguous uniform portion, the signal model is

$$\mathbf{z}_{-B} = \mathbf{A}_{B-} \Psi^* \mathbf{p} \quad (10)$$

(4) Virtual subarray at position $+2B$: This subarray originates solely from the sum co-array:

$$\mathcal{V}_{+2B} = \{(s_{1,i} + s_{1,j}) + 2B/d_1\}_{i,j=1}^M \quad (11)$$

With N_{2B+} elements in the maximum contiguous uniform portion, the signal model is

$$\mathbf{z}_{+2B} = \mathbf{A}_{2B+} \Psi^2 \mathbf{p} \quad (12)$$

(5) Virtual subarray at position $-2B$: Similarly,

$$\mathcal{V}_{-2B} = \{-(s_{1,i} + s_{1,j}) - 2B/d_1\}_{i,j=1}^M \quad (13)$$

The signal model is

$$\mathbf{z}_{-2B} = \mathbf{A}_{2B} \mathbf{\Psi}^* \mathbf{p} \quad (14)$$

In summary, through vectorization and rearrangement of the non-circular signal extended covariance matrix, the original $4N$ -element distributed nested array constructs five discrete virtual subarrays. The subarray at position 0 provides unambiguous but lower-resolution DOA estimation; the subarray pair at $\pm B$ has an equivalent $2B$ baseline, improving estimation accuracy but suffering from grating lobe ambiguity; the subarray pair at $\pm 2B$ has an equivalent $4B$ baseline, further enhancing accuracy but with more severe ambiguity problems. This hierarchical virtual array structure lays the foundation for the subsequent multi-level ambiguity resolution algorithm.

III. THREE-LEVEL AMBIGUITY RESOLUTION DOA ESTIMATION ALGORITHM

A. Spatial Smoothing and Covariance Matrix Reconstruction

For the virtual subarray signals extracted from the vectorized data, an improved spatial smoothing technique is employed to restore full-rank covariance matrices. Taking the virtual subarray at position 0 as an example, let its contiguous uniform portion length be N_0 , and select the smoothing subarray length $N_{s0} = \lfloor N_0/2 \rfloor$, then $K_0 = N_0 - N_{s0} + 1$ overlapping subarrays can be constructed. The forward-backward spatially smoothed covariance matrix is defined as

$$\mathbf{R}_0^{\text{FB}} = \frac{1}{2K_0} \sum_{k=1}^{K_0} \left(\mathbf{z}_0^{(k)} (\mathbf{z}_0^{(k)})^H + \mathbf{J} \mathbf{z}_0^{(k)*} (\mathbf{z}_0^{(k)*})^H \mathbf{J} \right) \quad (15)$$

where $\mathbf{z}_0^{(k)}$ is the signal vector of the k -th subarray, and $\mathbf{J}$ is the exchange matrix. For the virtual subarray pairs at $\pm B$ and $\pm 2B$, joint spatial smoothing is employed to exploit their conjugate symmetry, constructing joint covariance matrices $\mathbf{R}_B$ and $\mathbf{R}_{2B}$.

B. Three-Level Ambiguity Resolution Strategy

The weighted eigensubspace method [12] is employed for spectral estimation, whose spatial spectrum is given by

$$\hat{P}(\theta) = \frac{\|\mathbf{a}^H(\theta) \mathbf{U}_s \hat{\boldsymbol{\lambda}}\|^2}{\|\mathbf{a}^H(\theta) \mathbf{U}_n\|^2} \quad (16)$$

where $\mathbf{U}_s$ and $\mathbf{U}_n$ denote the signal and noise subspaces, respectively, and $\hat{\boldsymbol{\lambda}}$ is the weighting vector introduced in the signal subspace, defined as

$$\hat{\boldsymbol{\lambda}} = \left[\frac{1}{\lambda_1 - \hat{\sigma}^2}, \frac{1}{\lambda_2 - \hat{\sigma}^2}, \dots, \frac{1}{\lambda_P - \hat{\sigma}^2} \right]^T \quad (17)$$

with $\hat{\sigma}^2 = \frac{1}{N-P} \sum_{i=P+1}^{\bar{N}} \lambda_i$ being the estimated noise power. λ_i are the eigenvalues of the smoothed covariance matrix in descending order and $\bar{N}$ is the corresponding subarray dimension.

Level 1: Unambiguous coarse estimation. The virtual subarray at position 0 yields an unambiguous coarse estimate $\hat{\theta}_0^{(c)}$. The non-circular phase is estimated as

$$\hat{\phi}_p^{(c)} = \arg \left(\mathbf{a}_0^H(\hat{\theta}_0^{(c)}) \mathbf{U}_{s,0} \mathbf{U}_{s,0}^H \mathbf{a}_0^*(\hat{\theta}_0^{(c)}) \right) \quad (18)$$

Level 2: Short-baseline fine estimation. Based on $\hat{\theta}_0^{(c)}$, the search range for $\pm B$ subarrays is

$$\mathcal{Q}_{fs} = \left[\arcsin \left(\sin \hat{\theta}_0^{(c)} - \frac{\lambda}{2B} \right), \arcsin \left(\sin \hat{\theta}_0^{(c)} + \frac{\lambda}{2B} \right) \right] \quad (19)$$

The fine estimate $\hat{\theta}_0^{(fs)}$ is obtained within $\mathcal{Q}_{fs}$.

Level 3: Long-baseline fine estimation. Using $\hat{\theta}_0^{(fs)}$, the search range for $\pm 2B$ subarrays is

$$\mathcal{Q}_{fl} = \left[\arcsin \left(\sin \hat{\theta}_0^{(fs)} - \frac{\lambda}{4B} \right), \arcsin \left(\sin \hat{\theta}_0^{(fs)} + \frac{\lambda}{4B} \right) \right] \quad (20)$$

yielding the final high-accuracy estimate $\hat{\theta}_0^{(fl)}$.

C. Non-Circular Phase Assisted Ambiguity Resolution

As an intrinsic property of the source, the non-circular phase should remain consistent across estimations from different virtual subarrays. This property provides a supplementary criterion independent of the spatial spectrum for ambiguity resolution. The specific implementation is as follows:

For a candidate DOA estimate $\hat{\theta}$, extract the non-circular phase estimate from the signal subspace of the corresponding virtual subarray. Since the steering vector at the true DOA is highly correlated with the signal subspace, while the steering vector at grating lobe positions is affected by non-circular phase modulation in its correlation with the signal subspace, false peaks can be eliminated by comparing the non-circular phases from different level estimates.

Define the non-circular phase consistency cost function:

$$J(\hat{\theta}) = \sum_{k=1}^P |e^{j\hat{\phi}_k^{(\text{current})}} - e^{j\hat{\phi}_k^{(\text{ref})}}|^2 \quad (21)$$

where $\hat{\phi}_k^{(\text{current})}$ is the non-circular phase estimated at the current level, and $\hat{\phi}_k^{(\text{ref})}$ is the non-circular phase estimate from the reference level (previous level). When multiple local peaks exist within the search interval, select the peak that minimizes $J(\hat{\theta})$ as the correct estimate.

This three-level ambiguity resolution strategy combined with non-circular phase verification achieves progressive refinement from coarse to fine, fully exploiting the high-resolution advantage of long baselines while ensuring unambiguous estimation.

IV. SIMULATION RESULTS

In this section, numerical simulations are conducted to evaluate the performance of the proposed method.

Simulation 1: This experiment visualizes the spatial spectra at each estimation level. Each nested subarray contains $M = 8$ elements with baseline $B = 20\lambda$. Ten uncorrelated BPSK sources impinge from directions uniformly distributed

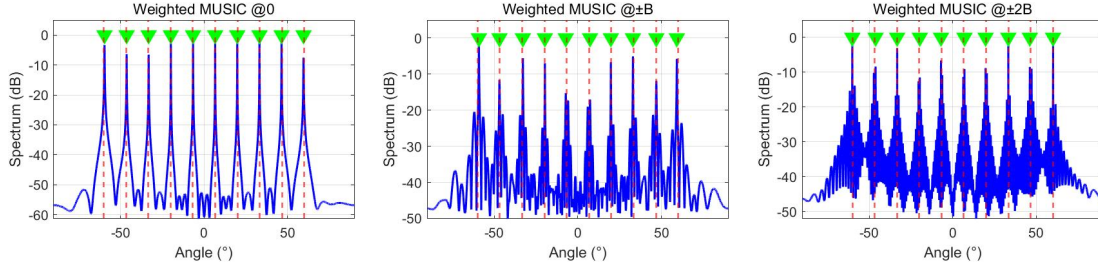


Fig. 2. Spatial spectra at three estimation levels.

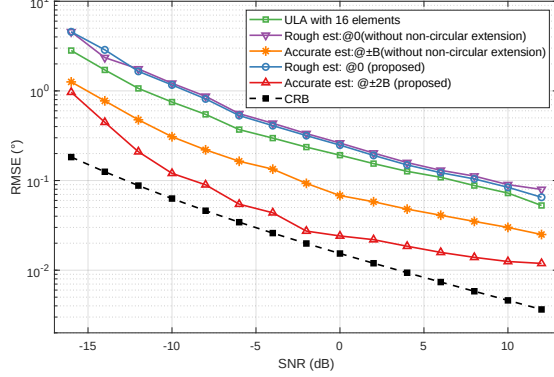


Fig. 3. RMSE versus SNR.

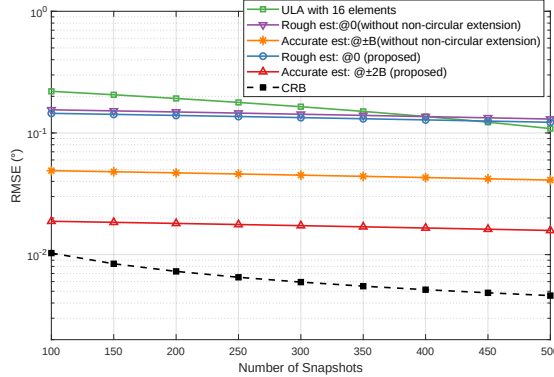


Fig. 4. RMSE versus number of snapshots.

in $(-60^\circ, 60^\circ)$ with non-circular phases $\phi = \pi/6$. The SNR is 10 dB with $L = 500$ snapshots.

As shown in Fig. 2, the Level 1 spectrum exhibits broad mainlobes (width $\approx 4.5^\circ$) due to limited virtual aperture. The Level 2 spectrum from $\pm B$ subarrays displays sharper peaks (width $\approx 1.1^\circ$) with grating lobes at intervals $\Delta\theta \approx \lambda/(2B)$. The Level 3 spectrum achieves the sharpest peaks (width $\approx 0.55^\circ$). The progressive sharpening validates the hierarchical strategy in exploiting long baselines while maintaining unambiguous estimation.

Simulation 2: The second experiment compares the RMSE performance. The array configuration remains identical to Simulation 1, with 500 snapshots and SNR varying from -16 dB to 12 dB. For each level, 500 Monte Carlo trials are

performed to compute the RMSE:

$$\text{RMSE} = \sqrt{\frac{1}{P \cdot N_{\text{MC}}} \sum_{n=1}^{N_{\text{MC}}} \sum_{p=1}^P (\hat{\theta}_p^{(n)} - \theta_p)^2} \quad (22)$$

where N_{MC} is the number of Monte Carlo trials.

Fig. 3 presents the RMSE comparison results versus SNR. At low SNR (below -10 dB), the Level 3 long-baseline estimation also yields a high RMSE due to estimation errors in the Level 1 coarse estimation. But still it maintains an estimation success rate of over 90% at $\text{SNR} = -16$ dB (with the deviation between estimated and true values less than one degree). As SNR increases, the proposed method with three-level fine estimation demonstrates the fastest convergence and achieves the lowest RMSE. At $\text{SNR} = 10$ dB, the proposed fine estimation achieves an RMSE of approximately 0.02° . The performance gap between methods with and without non-circular extension validates the effectiveness of exploiting non-circular signal properties.

Simulation 3: The last experiment investigates the RMSE versus snapshots with SNR fixed at 10 dB. As shown in Fig. 4, the estimation accuracy improves with increasing snapshots for all methods. The proposed method with three-level fine estimation consistently achieves the best performance across all snapshot numbers, demonstrating its practical applicability. Furthermore, the fine estimation methods consistently outperform their coarse counterparts, confirming that the hierarchical ambiguity resolution strategy successfully exploits the high-resolution capability of long baselines.

V. CONCLUSION

This paper has proposed a three-level ambiguity resolution DOA estimation method for distributed nested arrays based on non-circular signals. By exploiting the conjugate symmetry property of non-circular signals, an extended covariance matrix is constructed and vectorized to form five virtual subarrays located at positions $0, \pm B$, and $\pm 2B$. A hierarchical ambiguity resolution strategy is designed, where the virtual subarray at position 0 provides unambiguous coarse estimation, and the subarray pairs at $\pm B$ and $\pm 2B$ progressively refine the estimates while leveraging the non-circular phase consistency as an auxiliary criterion to eliminate false peaks. Simulation results demonstrate that the proposed method effectively exploits the long-baseline advantages of distributed arrays to achieve high-accuracy DOA estimation while ensuring unambiguous estimation.

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